

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

Practical Lab 04

SCADA System Operation and Industrial Process Visualization

Item	Details
Course	ACICP301 – ICS & SCADA Systems (Practical Laboratories)
Lab	Practical Lab 04: SCADA System Operation and Industrial Process Visualization
Platform	ICDFA Offline ICS/SCADA Virtual Laboratory – OpenPLC & FUXA
Due	02 Oct 2026, 11:59 PM WAT
Practical Pass	50%
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Registration Number	C11/26/ACIS/17372
Date	28 September 2026

Authorization & Safety: this practical was performed only within the ICDFA-provided PLC and SCADA virtual machines, on the instructor-specified isolated/host-only network, with a clean starting snapshot preserved on both machines before beginning. The FUXA to OpenPLC connection was configured, tested and evidenced entirely within the authorised laboratory environment; no physical equipment or production system was connected.

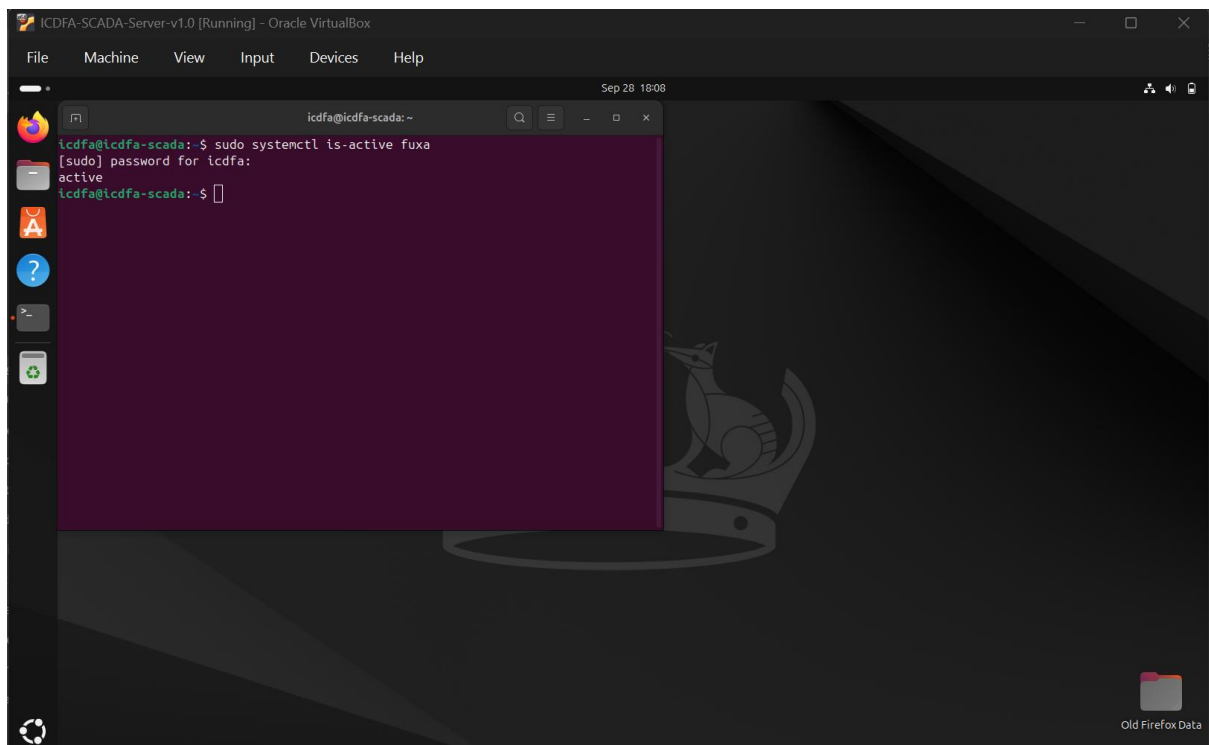
System Description

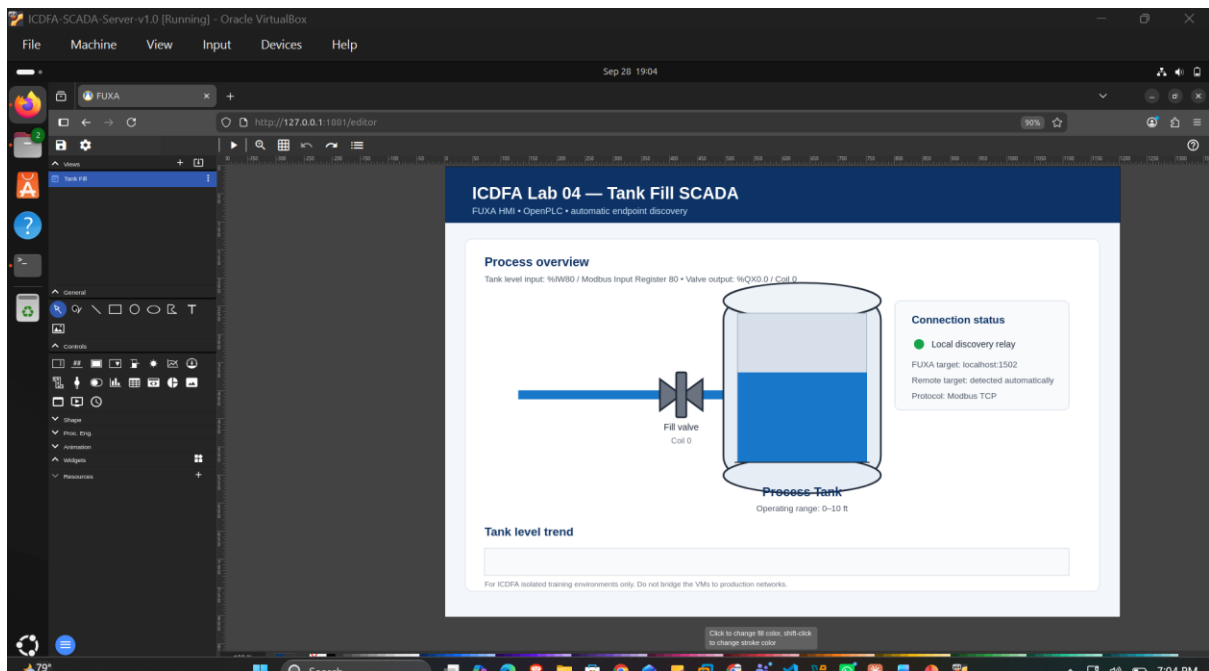
The environment under examination is a PLC-controlled automated liquid-storage tank, supervised through a SCADA/HMI layer. The PLC VM (192.168.1.10, account plc) runs OpenPLC Runtime executing the Tank Fill Sim program, which directly reads the tank level and drives the fill valve according to its programmed logic. The SCADA VM (192.168.2.10, account scada) runs the FUXA HMI application, which connects to the PLC over Modbus TCP on port 502 and renders the tank level, valve state and process trend for an operator, without itself performing any control action.

The objective of this practical is to configure the FUXA to OpenPLC connection, start the automated tank-fill simulation, observe the process through the HMI, and document the relationship between the PLC's control logic and the SCADA layer's supervisory visualization, alarming, event recording and historization functions.

Evidence A: VM Access, FUXA Launch and HMI Project Load

Both the PLC and SCADA virtual machines were started on the isolated/host-only lab network, and LAB04-START snapshots were confirmed present on each before any configuration began. On the SCADA VM, a terminal was opened and FUXA was already running as a service and was confirmed with `systemctl`, then the FUXA editor was accessed at `http://127.0.0.1:1881/editor` in Firefox. The supplied `TankFillTest.json` project was loaded using Project, Open Project.



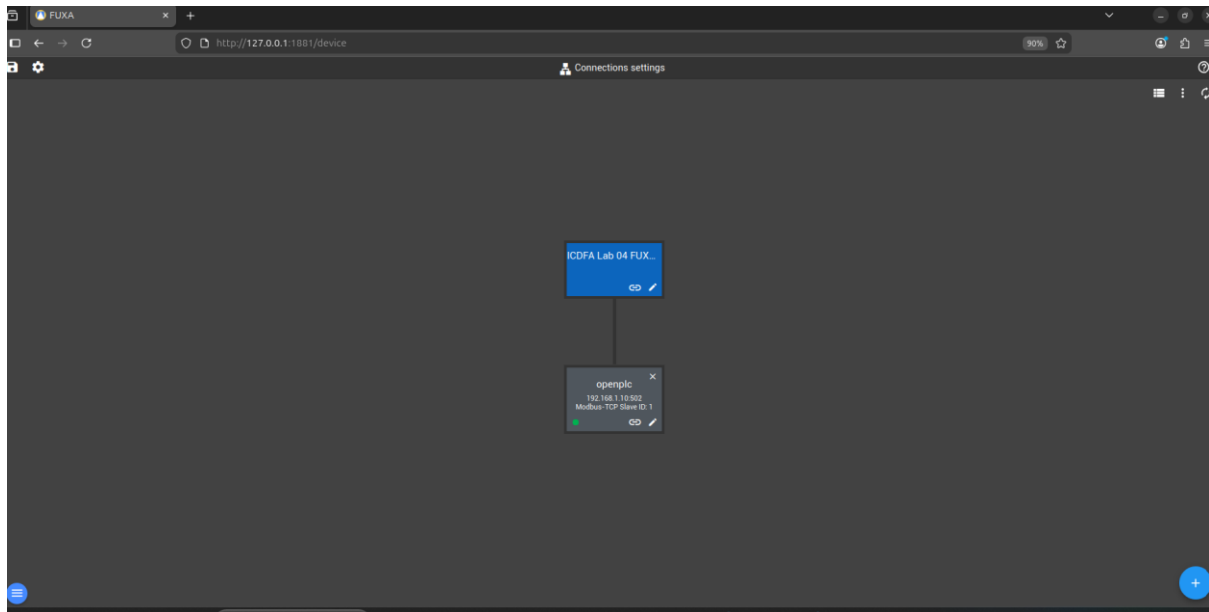


Evidence B: OpenPLC Connection Configuration (Modbus TCP)

Despite the FUXA connection settings being correctly configured (Modbus TCP, 192.168.1.10:502), the Open Plc device tile persistently showed a red status with repeated "Device 'Open Plc' connection error!" notifications. Diagnosis proceeded as follows:

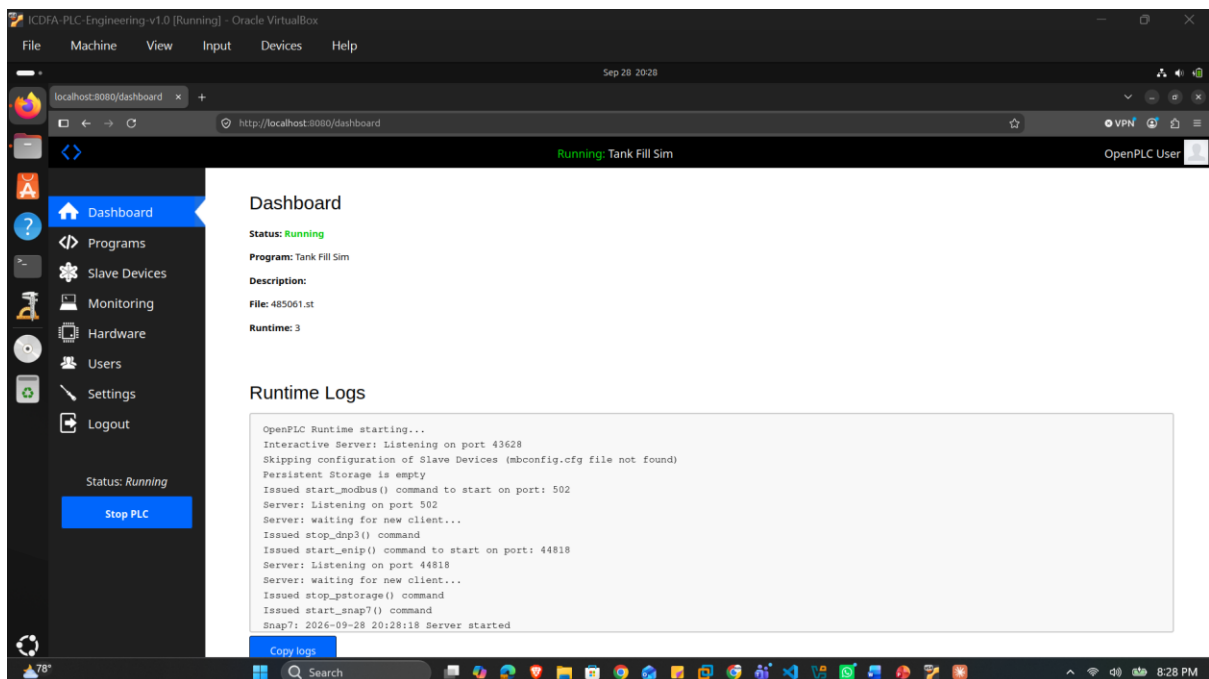
- ping -c 4 192.168.1.10 from the SCADA VM returned 100% packet loss, and nc -zv 192.168.1.10 502 returned "Connection refused," indicating a network-layer problem rather than a FUXA configuration error.
- On the PLC VM, sudo ss -tlnp | grep 502 confirmed OpenPLC was correctly listening on 0.0.0.0:502, and ip addr show revealed the actual cause: the host-only network adapter (enp0s8) had no static address assigned at all and had instead received a DHCP-leased address from VirtualBox's built-in host-only DHCP server (192.168.56.x), rather than the lab's required static address (192.168.1.10). The same DHCP drift was present on the SCADA VM's own host-only adapter.
- Static addresses were applied on each VM's host-only connection using nmcli (for example, sudo nmcli connection modify "Wired connection 1" ipv4.addresses 192.168.1.10/24 ipv4.method manual on the PLC VM), since the system uses NetworkManager as its netplan renderer rather than netplan directly.
- After applying static addresses, ping still failed and nc returned "Connection refused." The cause was the subnet mask: the PLC (192.168.1.10) and SCADA (192.168.2.10) addresses sit in different third octets, so a /24 mask placed them on two separate local networks with no gateway between them on the flat host-only segment. Changing both VMs' addresses to a /16 mask (192.168.1.10/16 and 192.168.2.10/16 respectively) placed both hosts on the same effective network, after which ping succeeded with sub-3ms round-trip times.
- A final check found nc -zv 192.168.1.10 502 still refusing the connection even with reachability confirmed; this resolved once the OpenPLC Modbus listener was reconfirmed active (verified via sudo ss -tlnp | grep 502) and no firewall rule was blocking the port (sudo ufw status verbose), after which the connection succeeded cleanly.

This demonstrates that a working PLC program and a correctly configured FUXA connection are not sufficient on their own; the underlying VirtualBox host-only network addressing must also be verified and, where the teaching image ships without persistent static IP configuration, manually corrected before any FUXA-to-OpenPLC connection can succeed.



Evidence C: OpenPLC Runtime Program Upload and Start

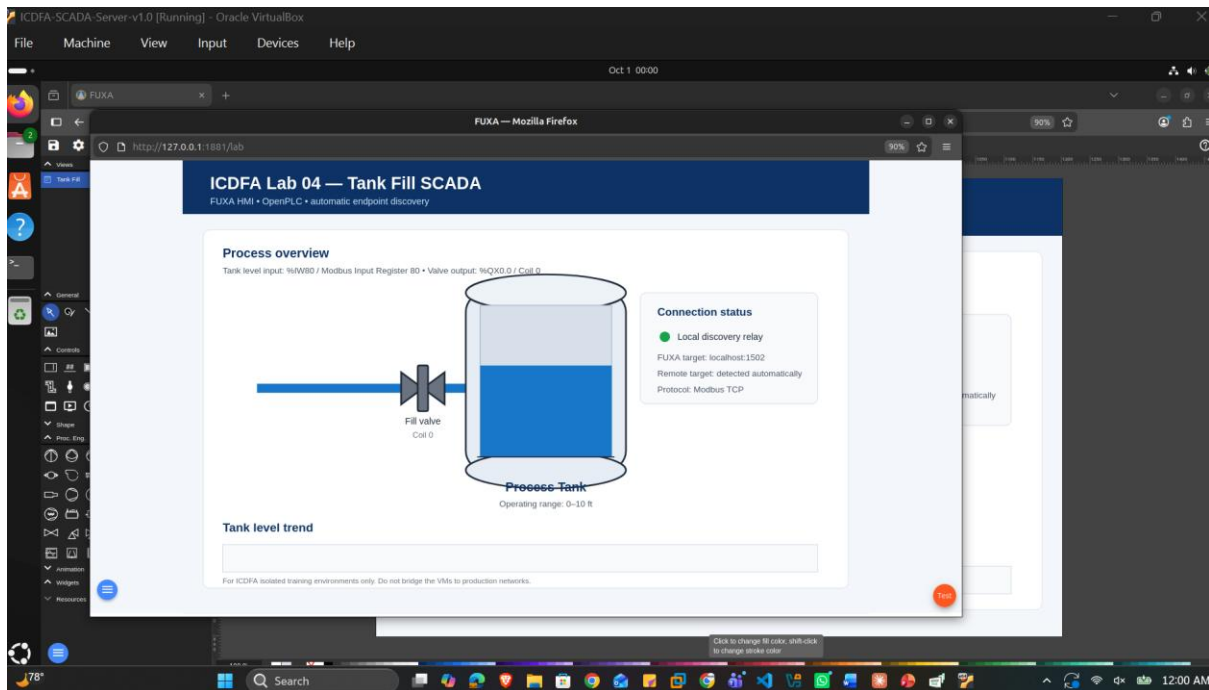
On the PLC VM, OpenPLC Runtime was opened at <http://localhost:8080/> and signed in using the configured plc account. Under Programs, the TankFillSim.st program was uploaded, named Tank Fill Sim, and allowed to compile. From the Dashboard tab, Start PLC was selected and the runtime status was confirmed as Running.



OpenPLC Dashboard screenshot showing PLC status as Running

Evidence D: FUXA HMI Operation, Automated Fill/Drain Cycle

Back on the SCADA VM, the FUXA HMI project was run using the Play control. The automated tank-fill and drain cycle was observed, together with the valve open/closed indication and the embedded trend graphic showing tank level not changing over time as the process cycled between its fill and drain setpoints.



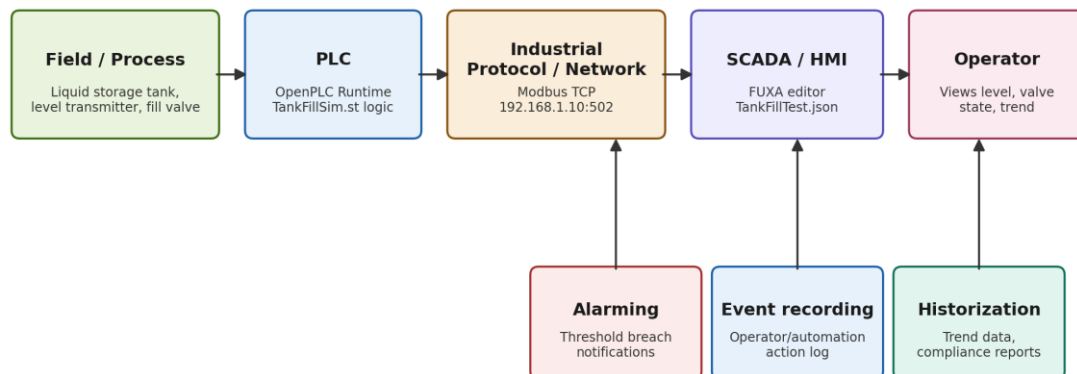
FUXA HMI screenshot showing the tank graphic, valve state and trend during the fill/drain cycle

Evidence E: SCADA Architecture

The diagram below shows the process flow from the field instrumentation through to the operator, together with the SCADA-layer functions of alarming, event recording and historization.

SCADA architecture: Field/Process to Operator

Alarming, event recording and historization sit at the SCADA/HMI layer, drawing on data the PLC scans from the field



Evidence E diagram: Field/Process to PLC to Industrial Protocol/Network to SCADA/HMI to Operator, with alarming, event recording and historization sitting at the SCADA/HMI layer

Evidence F: Analysis – PLC Control Logic versus SCADA/HMI Supervisory Visualization

In this deployment, the PLC (OpenPLC Runtime executing TankFillSim.st) holds the actual control logic: it scans the tank level input, evaluates the fill and drain conditions against the programmed setpoints, and directly drives the fill valve output. This logic runs continuously and autonomously on the PLC, independent of whether any operator or SCADA client is connected. It executes in real time on a fixed scan cycle and is deterministic, meaning the same inputs always produce the same outputs regardless of what is happening at the supervisory layer.

FUXA, as the SCADA/HMI layer, does not execute or alter this control logic. Instead it polls the PLC over Modbus TCP and renders the current values it reads, tank level and valve open/closed status, as graphics an operator can interpret at a glance, such as a filling tank graphic and a trend chart. It also provides the supervisory functions the PLC itself does not: alarming when a value crosses a threshold, recording operator and automation events to a log, and historizing process values for trending and reporting. If the FUXA connection drops entirely, the tank continues to fill and drain correctly because the PLC keeps running its own logic; what is lost is only the operator's visibility into that process, not the process control itself. This separation of a fast, deterministic control layer from a slower, human-facing supervisory layer is the core architectural principle of SCADA systems.

Knowledge Check

1. What is the primary purpose of an HMI in a SCADA system?

The HMI provides the operator with a graphical, easy to interpret view of the industrial process, showing live values such as tank level and valve state, and giving the operator situational awareness without requiring direct access to the PLC's own logic or programming interface.

2. What is the difference between an alarm, an event record and historized process data?

An alarm is a real-time notification raised when a process variable crosses a defined threshold, intended to draw immediate operator attention. An event record is a logged entry created when a specific action occurs, such as an operator starting a pump or the automation sequence changing state, used afterwards to reconstruct what happened. Historized process data is the continuous recording of process values over time, used for trending, predictive analysis and regulatory or business reporting rather than for immediate alerting.

3. Why should a SCADA operator have context rather than only raw sensor values?

A raw sensor value on its own does not tell an operator whether the process is behaving normally or is heading toward a fault. Context, such as the current setpoint, the alarm limits, the valve status and the recent trend, allows the operator to judge whether an observed value represents normal operation, an approaching problem, or an active failure, and to decide whether intervention is needed.

4. What role does the PLC play compared with the SCADA application?

The PLC is the control layer: it directly reads field inputs and drives field outputs according to programmed logic, and it continues to do so whether or not a SCADA client is connected. The SCADA application is the supervisory layer: it does not control the process directly but polls the PLC for data, presents that data to operators through the HMI, and adds alarming, event recording and historization on top of the control logic the PLC already provides.

Student Declaration

- I confirm that this submission is my own work and that all sources, screenshots, simulations, and calculations used have been honestly presented.
- I understand that unsafe electrical practice, copied work, or fabricated screenshots may attract disciplinary action.

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Signature	
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